LAB 7: FUZZING FOR INPUT VALIDATION BUGS

Software Engineering Laboratory

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Tools: Python 3.13.7, Hypothesis 6.170.15, pytest 9.2.0

Objective

To use property-based fuzz testing with the Hypothesis library to identify input-validation failures in three Python functions, then harden the functions and rerun the tests.

Functions tested

1. **verify_isbn10_format()** — handles ISBN-10 input.
2. **calculate_depreciation_factor()** — calculates a book depreciation factor from age/lifespan input.
3. **lookup_condition_multiplier()** — maps book-condition codes to multipliers.

Execution summary

The original application executed successfully with the sample values ISBN **0-306-40615-2**, age/lifespan **3/10**, and condition **MINT**. Hypothesis then exposed failures in all three functions. After defensive input handling was added, the same three fuzz tests passed.

Evidence screenshots

The following screenshots document the original run, fuzz-discovered failures, successful regression run, and fixed application run.

Evidence 1

```
PS D:\SE LAB\L6_L7_09_bookstore_trade_in> Get-Content ".\Lab 7\test_processor.py"
@given(st.text() | st.none())
def test_calculate_depreciation_factor_safe(s):
    # To-Do:Using try and except complete this function.
    pass

@given(st.text() | st.none())
def test_lookup_condition_multiplier_safe(s):
    # To-Do:Using try and except complete this function.
    pass
PS D:\SE LAB\L6_L7_09_bookstore_trade_in> python ".\Lab 7\processor.py"
==== Running verify_isbn10_format ====
Enter ISBN-10 (e.g. 0-306-40615-2): 0-306-40615-2
Group 306: Starts 0, Checksum 2

==== Running calculate_depreciation_factor ====
Enter age/lifespan in years (e.g. 3/10): 3/10
Depreciation Factor: 0.7

==== Running lookup_condition_multiplier ====
Enter condition (e.g. MINT, GOOD, POOR): MINT
Multiplier: 1.25
PS D:\SE LAB\L6_L7_09_bookstore_trade_in>
```

Figure 1. Original application execution with valid sample inputs.

Evidence 2

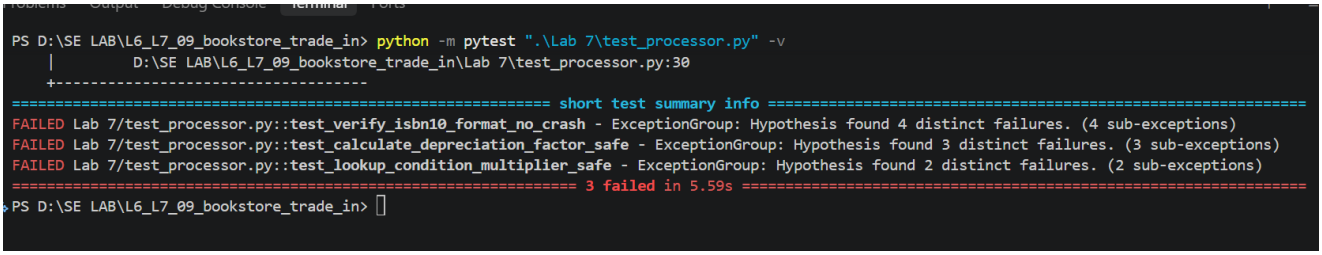


Figure 2. Hypothesis fuzz tests exposing failures in all three functions.

Evidence 3

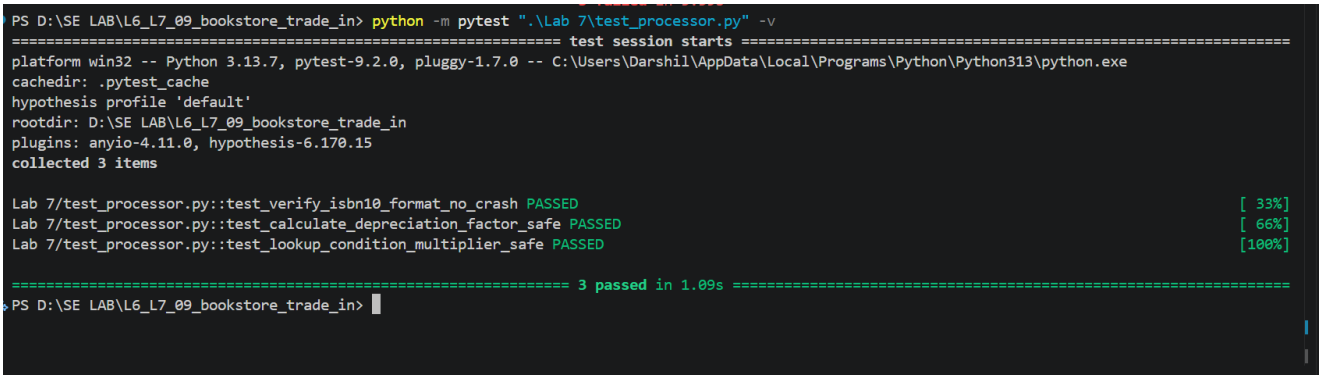


Figure 3. Re-run after fixes: all three Hypothesis tests passed.

Evidence 4

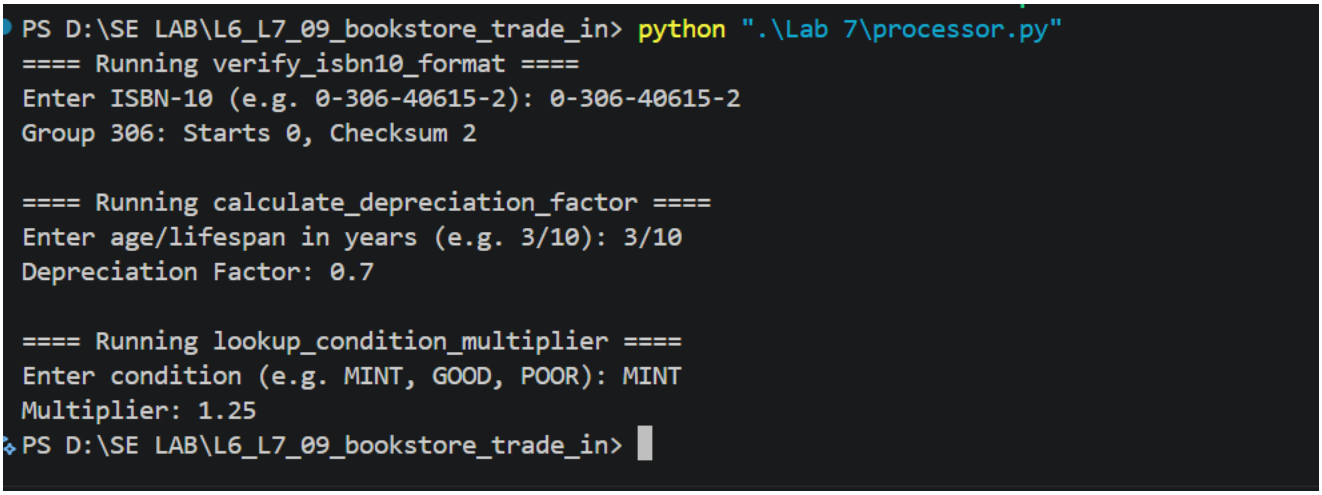


Figure 4. Fixed application execution with the same valid sample inputs.